Haowen Xu, Ph.D., SMIEEE

CONTACT INFORMATION

Research Fellow (Level B)
School of Built Environment
UNSW Sydney
Kensington, NSW, Australia

Former Adjunct Associate Professor, University of Tennessee, Knoxville
Research Scientist, Oak Ridge National Laboratory, U.S. Department of Energy

Work Email: haowen.xu.phd@gmail.com
Official Email: haowen.xu1@unsw.edu.au

WEB PRESENCE Personal Website – Google Scholar – LinkedIn – ORCID

RESEARCH EXPERTISE

Focus Areas:

- Intelligent Transportation Systems (ITS), and Connected and Automated Vehicles (CAVs).
- Generative and Agentic AI for autonomous decision-making, workflow automation, and surrogate modelling.
- AI-enabled Digital Twins for transportation, infrastructure, urban, and environmental systems.
- Transportation Network Modelling and Optimization for multimodal transportation systems, freight logistics, routing optimization, and infrastructure decision support.
- Cyber-Physical Systems for V2I-enabled intelligent transportation, connected infrastructure, advanced driver-assistance systems, and traffic signal control.
- AI-enabled Knowledge Graphs and Ontologies for data integration, semantic reasoning, and urban decision-support systems.
- 3D Spatial Computing, GIS, High-Performance Computing, and Urban Informatics for large-scale modelling and simulation of urban, environmental, and transportation systems.
- Immersive Visual Analytics, including VR/AR and Human-AI Interaction.

Technical Skills

Programming: Python, C/C++, JavaScript, TypeScript, SQL
Artificial Intelligence & Machine Learning: PyTorch, TensorFlow, Scikit-learn
Agentic AI & Large Language Models: Ollama, vLLM, Model Context Protocol (MCP)
Traffic & Transportation Simulation: PTV Vissim, SUMO, CARLA
Scientific Computing: High-Performance Computing, CUDA, Taichi, GPU Computing
Spatial Analytics & GIS: ArcGIS, QGIS, GDAL
Operations Research & Simulation: Gurobi, AnyLogic (Agent-Based Simulation)
Digital Twin Platforms & Software Development: FastAPI, Django, Node.js
Knowledge Graphs & Data Infrastructure: Neo4j, PostgreSQL, MongoDB
Cloud & DevOps: Docker, Kubernetes, Git
Scientific Visualization & Immersive Technologies: D3.js, WebGL, VTK, Unity, Unreal Engine

RESEARCH PROFILE

- Seven years of post-Ph.D. research experience spanning intelligent transportation systems (ITS), AI-enabled digital twins, scientific machine learning, transportation network modelling, cyber-physical systems, 3D spatial computing, and urban informatics.
- Author of **30+ peer-reviewed journal articles and book chapters** and **30+ conference publications, technical reports, and white papers**, including **10+ first- or corresponding-**

author papers in leading journals such as *Computers, Environment and Urban Systems*, *IEEE Transactions on Intelligent Transportation Systems*, *IEEE Transactions on Intelligent Vehicles*, *Transportation Research Record, Science of the Total Environment*, and *Environmental Modelling & Software.*

- **1,600+ citations**, **h-index 21** (Google Scholar), and **IEEE Senior Member**.
- **Co-Principal Investigator (Co-PI)** on competitive research projects funded by the U.S. Department of Energy (USDOE), including ARPA-E and the Nuclear Safety Research and Development (NSRD) Program, and Principal Investigator (PI) of the 2026 UNSW Arts, Design & Architecture (ADA) Faculty Research Grant.
- **Recipient** of **competitive research awards** from international professional societies, including **IEEE**, International Conference on Computers and Industrial Engineering (CIE), the **International Building Performance Simulation Association (IBPSA)**, and the **American Water Resources Association (AWRA)**.
- Inventor of **3 U.S. patent applications** in Intelligent Transportation Systems and Connected Vehicle technologies (Application Number - US20250131819A1, US20250209911A1 & US20250128708A1).
- **Adjunct Associate Professor** at the **University of Tennessee, Knoxville (UTK)**, with experience supervising and co-supervising Ph.D. students.
- **Australian Permanent Resident (PR)** based in Sydney and an **early career researcher**, with 1.5 years of full-time equivalent (FTE) university academic employment as a Research Fellow at UNSW Sydney.

EDUCATION

Ph.D., Civil Engineering 2015 - 2019
The University of Iowa, Iowa City, IA
Cumulative GPA: 3.90/4.00
Dissertation Title: "Data-Driven Framework for Forecasting Sedimentation at Culverts"
Focus Area: Hydroinformatics and Environmental Informatics
Passed the final thesis defense (on December 5th, 2018)

- Received multiple travel grants and research fellowship from NSF, UCGIS, and CGRER
- UI SETP (the University of Iowa Student Employee of the Year) Nominee

Graduate Certificate in Geoinformatics 2014 - 2016
The University of Iowa, Iowa City, IA
Cumulative GPA: 3.94/4.00

MS, Civil and Environmental Engineering 2013 - 2015
The University of Iowa, Iowa City, IA
Cumulative GPA: 4.00/4.00
Dissertation Title: "Prototype Hydroinformatics-based System for Supporting Decision Making in Culvert Design and Monitoring"

- Received a Full-tuition Scholarship

B.S.E., Civil Engineering 2009 - 2013
The University of Iowa, Iowa City, IA
Cumulative G.P.A.: 3.54/4.00; Dean's List
Thesis Title: "Search Engine for Iowa Watershed Data Information System"

- *Graduated with Honors* in Civil Engineering

PROFESSIONAL EXPERIENCE

UNSW Sydney March 2025 - Present
Research Fellow (Level B)
School of Built Environment

Oak Ridge National Laboratory, US Department of Energy June 2021 – March 2025
Research Scientist, Computational Urban Sciences Group
Computational Sciences and Engineering Division

University of Tennessee, Knoxville September 2024 - July 2026
Adjunct Associate Professor
Department of Industrial and Systems Engineering (ISE)

Oak Ridge National Laboratory, US Department of Energy June 2019 - May 2021
Postdoctoral Research Associate, Computational Urban Sciences Group
Computational Sciences and Engineering Division

University of Iowa August 2013 - May 2019
Graduate Research Assistant
IIHR – Hydroscience & Engineering, Iowa City, Iowa

Institute of Rural Research and Development (IRRAD) December 2013 - January 2014
Teaching Assistant & Engineering Surveyor & GIS Specialist
Gurgaon, India

HONORS AND AWARDS

2024 Best Research Paper Award from the 51st International Conference on Computers & Industrial Engineering (CIE51), UNSW, Australia

2022 Best Poster Award from the 19th IEEE International Conference on Mobile Ad-hoc and Smart Systems (MASS)

2022 Invited Presenter, U.S. National Academies (NASEM)

2022 Award for the Most Creative Demonstration from IBPSA HackSimBuild 2022 competition in conjunction with Building Performance Analysis Conference and SimBuild 2022

2021 Invited presenter at the 1st GeoScale Workshop

2020 Significant Event Award (SEA) – ORNL, USDOE
Release of Real-time Ctwin v1.1 -- The Chattanooga Digital Twin for Mobility/Transportation

2018 Wayne L. Paulson Scholarship Award (1000 USD)

2018 Graduate College Post-Comprehensive Research Fellowship (9000 USD)

2017 UCGIS & NSF Travel Awards (2000 USD)

2017 CGRER Travel Award (600 USD)

2017 UI SETP (the University of Iowa Student Employee of the Year) Nominee

2016 Invited presenter at the United States Environmental Protection Agency (USEPA) headquarters

2016 Best Student Oral Presentation Award from the American Water Resources Association (AWRA)

2015 CGRER Travel Award (750 USD)

2013 Study Abroad International Student Merit-based Scholarship

2013 Segal Family Foundation Scholarship

2013 The University of Iowa Engineering College Merit-based Scholarship

PROFESSIONAL MEMBERSHIP

- Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
- Association for Computing Machinery (ACM)
- Association of American Geographers (AAG)
- Chi-Epsilon National Civil Engineering Honor Society
- International Association for Hydro-Environment Engineering and Research (IAHR) Web Master of SIIHR and Iowa Young Professionals Network

JOURNAL PUBLICATION & Book Chapters

1. Yuan, J., Li, W., Wang, C., LaClair, T., **Xu, H.,** Berres, A., Kadav, P., Wang, H., Chen, B., Lim, H., Groelke, B., Shao, Y., Ekti, A. R., Chen, J., Wang, Q., & Sanyal, J. *Real-world integration of traffic signal and connected automated vehicle speed control for energy-efficient mobility*. *Nature Communications (in-principle accepted; under final editorial revision).*
2. **Xu, H.,** Zlatanova, S., Li, X., Wachowicz, M., & Batty, M. (2026). Towards fully automated city operations: integrating agentic AI with urban digital twins. *Computers, Environment and Urban Systems*, *128*, 102449.
3. **Xu, H.,** Lu, Q., Li, X., Liu, Z., Yu, X. Y., & Zhu, L. (2026). Bridging Industrial Engineering and Agentic AI: Integrating Operations Research Methods for Uncertainty-aware and Verifiable Spatial Decision Support. *Available at SSRN 7229001*.
4. **Xu H.,** Zlatanova, S., Liang, R., Canbulat, I., Heat-Transfer-Driven Voxel-Based Simulation: An Exploratory GPU-Accelerated Framework for Urban-Scale 3D Fire Spread. *ISPRS International Journal of Geo-Information*. 2026; 15(9):423. https://doi.org/10.3390/ijgi15090423
5. **Xu, H.,** Zlatanova, S., Gorte, B., Lamsal, R., Heslop, D., Liang, R., & Canbulat, I. (2026). A Pilot Study on a GPU-Accelerated Voxel Simulation Framework for 3D Indoor and Urban-Scale Gas Dispersion and Aerosol Transport. *ISPRS International Journal of Geo-Information*. 2026; 15(9):405. https://doi.org/10.3390/ijgi15090405
6. **Xu, H.,** Yuan, J., Li, W., Zhang, F., Zlatanova, S., Ban, X., ... & Ye, X. (2025). Next-Generation Intelligent Transportation Systems using Multimodal Generative AI and Diverse Modalities of Traffic Data. Handbook Of Urban Technologies (*accepted*), *Available at SSRN 5547519*.
7. **Xu, H.,** Zlatanova, S., Liang, R., & Canbulat, I. (2026). A Modular Light-weight Voxel-Based 3D Wildfire Propagation Simulator in Python Using LiDAR Data, High-Performance Computing (HPC), and Immersive Scientific Visualization. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *10*, 127-134.
8. **Xu, H.,** Zlatanova, S., Liang, R., & Canbulat, I. (2025). Generative AI as a Pillar for Predicting 2D and 3D Wildfire Spread: Beyond Physics-Based Models and Traditional Deep Learning. Fire, *8*(8), 293.
9. **Xu, H.,** Sun, Y., Tupayachi, J., Omitaomu, O., Zlatanova, S., & Li, X. (2025). Towards the Autonomous Optimization of Urban Logistics: Training Generative AI with Scientific Tools via Agentic Digital Twins and Model Context Protocol. *arXiv preprint arXiv:2506.13068*.

10. **Xu, H.,** & Yu, X. Y. (2025). From powerpoint ui sketches to web-based applications: Pattern-driven code generation for GIS dashboard development using knowledge-augmented llms, context-aware visual prompting, and the react framework. *arXiv preprint arXiv:2502.08756*.
11. **Xu, H.,** Yuan, J., Zhou, A., Xu, G., Li, W., & Ye, X. (2025). GenAI-powered Multi-Agent Paradigm for Smart Urban Mobility: Opportunities and Challenges for Integrating Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) with Intelligent Transportation Systems. Urban Human Mobility, 123-137.
12. **Xu, H.,** Boyaci, A., Lian, J., & Wilson, A. (2025). Explainable ai for multivariate time series pattern exploration: Latent space visual analytics with temporal fusion transformer and variational autoencoders in power grid event diagnosis. IEEE Access.
13. **Xu, H.,** Omitaomu, F., Sabri, S., Zlatanova, S., Li, X., & Song, Y. (2024). Leveraging generative AI for urban digital twins: a scoping review on the autonomous generation of urban data, scenarios, designs, and 3D city models for smart city advancement. Urban Informatics, 3(1), 29.
14. Tupayachi, J., **Xu, H*.,** Omitaomu, O. A., Camur, M. C., Sharmin, A., & Li, X. (2024). Towards Next-Generation Urban Decision Support Systems through AI-Powered Construction of Scientific Ontology Using Large Language Models—A Case in Optimizing Intermodal Freight Transportation. Smart Cities, 7(5), 2392-2421.
15. Moriano, P., Berres, A., **Xu, H.,** & Sanyal, J. (2024). Spatiotemporal Features of Traffic Help Reduce Automatic Accident Detection Time. Expert Systems With Applications, 244, 122813.
16. Shi, L., Qu, M., Liu, L., Venegas, T. P., Wang, L., Dong, J., Cui, B., **Xu, H.,** Liu, X., & Li Y. (2024). Performance evaluation of underground thermal storage integrated dual-source heat pump systems. Energy and Buildings: 114349.
17. Berres, A., Moriano, P., **Xu, H.,** Tennille, S., Smith, L., Storey, J., & Sanyal, J. (2024). A traffic accident dataset for Chattanooga, Tennessee. Data in brief, 55, 110675.
18. **Xu, H.,** Shao, Y., Chen, J., Wang, C., & Berres, A. (2023). Semi-Automatic Geographic Information System Framework for Creating Photo-Realistic Digital Twin Cities to Support Autonomous Driving Research. Transportation Research Record, 03611981231205884.
19. **Xu, H.,** Yuan, J., Berres, A., Shao, Y., Wang, C., Li, W., Laclair, T., Sanyal, J., & Wang, H. (2023). A Mobile Edge Computing Framework for Traffic Optimization At Urban Intersections Through Cyber-Physical Integration. IEEE Transactions on Intelligent Vehicles, doi: 10.1109/TIV.2023.3332256.
20. **Xu, H.,** Berres, A., Yoginath, S. B., Sorensen, H., Nugent, P. J., Severino, J., ... & Sanyal, J. (2023). Smart Mobility in the Cloud: Enabling Real-Time Situational Awareness and Cyber-Physical Control Through a Digital Twin for Traffic. IEEE Transactions on Intelligent Transportation Systems.
21. **Xu, H.,** Berres, A. S., Shao, Y., Wang, C., J, New., Omitaomu, O. A. (2022). Towards a Smart Metaverse City: Immersive Realism and 3D Visualization of Digital Twin Cities. In Surya Durbha (Eds.). Advances in Scalable and Geospatial Analytics: New Trends, Challenges and Applications.
22. Muste, M., & **Xu, H*.** (2022). Multi-pronged approach for monitoring sedimentation processes at multi-barrel culverts. Journal of Hydrology, 610, 127840.
23. **Xu, H.,** Berres, A., Liu, Y., Allen-Dumas, M. R., & Sanyal, J. (2022). An overview of visualization and visual analytics applications in water resources

management. Environmental Modelling & Software, 105396.

24. **Xu, H.,** Berres, A., Thakur, G., Sanyal, J., & Chinthavali, S. (2021). EPIsembleVis: A geo-visual analysis and comparison of the prediction ensembles of multiple COVID-19 models. Journal of Biomedical Informatics, 124, 103941.
25. **Xu, H.,** Berres, A., Tennille, S. A., Ravulaparthy, S. K., Wang, C., & Sanyal, J. (2021). Continuous Emulation and Multiscale Visualization of Traffic Flow Using Stationary Roadside Sensor Data. IEEE Transactions on Intelligent Transportation Systems.
26. **Xu, H.,** Wang, C., Berres, A., LaClair, T., & Sanyal, J. (2022). Interactive web application for traffic simulation data management and visualization. Transportation Research Record, 2676(1), 274-292.
27. Li, X., **Xu, H.,** Huang, X., Guo, C. A., Kang, Y., & Ye, X. (2021). Emerging geo-data sources to reveal human mobility dynamics during COVID-19 pandemic: opportunities and challenges. Computational Urban Science, 1(1), 1-9.
28. Berres, A. S., **Xu, H.,** Tennille, S. A., Severino, J., Ravulaparthy, S., & Sanyal, J. (2021). Explorative visualization for traffic safety using adaptive study areas. Transportation Research Record, 2675(6), 51-69.
29. Allen-Dumas, M. R., **Xu, H*.,** Kurte, K. R., & Rastogi, D. (2021). Toward urban water security: broadening the use of machine learning methods for mitigating urban water hazards. Frontiers in Water, 2, 562304.
30. Berres, A. S., LaClair, T. J., Wang, C., **Xu, H.**, Ravulaparthy, S., Todd, A., ... & Sanyal, J. (2020). Multiscale and Multivariate Transportation System Visualization for Shopping District Traffic and Regional Traffic. Transportation Research Record, 0361198120970526.
31. **Xu, H.,** Demir, I., Koylu, C., & Muste, M. (2019). A web-based geovisual analytics platform for identifying potential contributors to culvert sedimentation. Science of the Total Environment, 692, 806-817.
32. **Xu, H.,** Windsor, M., Muste, M., & Demir, I. (2020). A web-based decision support system for collaborative mitigation of multiple water-related hazards using serious gaming. Journal of Environmental Management, 255, 109887.
33. **Xu, H.,** Muste, M., & Demir, I. (2019). Web-based geospatial platform for the analysis and forecasting of sedimentation at culverts. Journal of Hydroinformatics, 21(6), 1064-1081.
34. Basnet, K., Muste, M., Constantinescu, G., Ho, H., & **Xu, H.** (2016). Close range photogrammetry for dynamically tracking drifted snow deposition. Cold Regions Science and Technology, 121, 141-153.

GRANT SUPPORTS

1. 2026. UNSW Arts, Design & Architecture (ADA) Faculty Research Grant: AUD $7,500
   Title: Faculty Research Grant – Development of Competitive External Research Funding Proposals
   Role: Principal Investigator (PI)
   Duration: 2026
   Award: Competitive internal research grant supporting the development of external competitive funding applications.
2. United States Department of Energy - Advanced Research Projects Agency-Energy (ARPA-E): $1,900,000
   Title: A Cognitive Freight Transportation Digital Twin for Resiliency and Emission Control

Through Optimizing Intermodal Logistics (RECOIL)

Role: Co-Principal Investigator (Co-PI)
Duration: Fiscal Year 2023-Present
Project URL: https://recoil.utk.edu/
URL: https://arpa-e.energy.gov/technologies/exploratory-topics/intermodal-freight

3. United States Department of Energy - Nuclear Safety Research and Development (NSRD): $500,000
Title: ML-Assisted Atmospheric Hazard Modeling for Effective Geospatial Risk Analysis
Role: Co-Investigator
Duration: Fiscal Year 2023-Present

4. United States Department of Energy - Vehicle Technologies Office (VTO): $4,000,000
Title: Scaling up the Realtime Data, Simulation and Artificial Intelligence (AI) and Control for Optimizing Regional Mobility
Role: Task Lead for developing Digital Twins
Duration: Fiscal Year 2021-2023
URL: https://www.energy.gov/eere/vehicles/articles/regional-mobility-chattanooga

5. The Iowa Department of Transportation - The Iowa Highway Research Board (IHRB)
Title: Forecasting and Mitigation of Sedimentation at Culverts
Role: Co-Investigator
Duration: 01/2015-12/2015
URL:
https://publications.iowa.gov/27032/1/TR%20665_Final%20Report_Mitigation%20of%20Sedimentation%20at%20Multi-Box%20Culverts.pdf

REFEREED CONFERENCE PROCEEDINGS

1. Xu, H., Zlatanova, S., Liang, R., & Canbulat, I. (2026). An Exploratory Study of Transformer-Based Generative AI Surrogate Modeling for 3D Wildfire Spread Simulation in Voxelized City. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *12*, 317-324.

2. Xu, H., Zlatanova, S., Liang, R., & Canbulat, I. (2026). Community-oriented Wildfire Planning with Agentic AI: Automated Fire Spread Simulation and Interactive 3D Visualization. *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *11*, 251-258.

3. Yoginath, S. B., Ahmad, N., Gunaratne, C., Amichi, L., Kim, J. S., Burger, A., ... & Thakur, G. M. (2025, November). A Scalable Multi-Modal Framework for High-Fidelity Distributed Human Mobility Simulations. In *Proceedings of the 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation* (pp. 1-11).

4. Li, X., **Xu, H.,** Tupayachi, J., Omitaomu, O., & Wang, X. (2024). Empowering Cognitive Digital Twins with Generative Foundation Models: Developing a Low-Carbon Integrated Freight Transportation System. *The* 51*st International Conference on Computers & Industrial Engineering (*CIE51*),* December 11th, Sydney, NSW, Australia. arXiv preprint arXiv:2410.18089.

5. **Xu, H.,** Li, X., Tupayachi, J., Lian, J., & Omitaomu, O. A. (2024). Automating bibliometric analysis with sentence transformers and retrieval-augmented generation (RAG): A pilot study in semantic and contextual search for customized literature characterization for high-impact urban research. *In The 2nd ACM SIGSPATIAL International Workshop on Advances in Urban AI*, November 1, 2024, Atlanta, GA, USA. ACM.

6. **Xu, H.,** Liu, X., Dong, J., & Lian, J. (2024). A Digital Twin Approach to Support Real-time Situational Awareness and Intelligent Cyber-physical Control in Energy Smart Buildings. In Digital Innovation Center Of Excellence (DICE) - Idaho National Laboratory (INL), Idaho Falls, Idaho, May 1, 2024.
7. Niloofar, P., Lazarova-Molnar, S., Omitaomu. O., **Xu, H.,** & Li, X. (2023). A General Framework for Human-in-the-loop Cognitive Digital Twins. In 2023 Winter Simulation Conference. San Antonio, Texas, December 10-13, 2023.
8. Shao, Y., Wang, C., Berres, A., Yoshioka, J., Cook, A., & **Xu, H.** (2022). Computer vision-enabled smart traffic monitoring for sustainable transportation management. In International Conference on Transportation and Development 2022 (pp. 34-45).
9. **Xu, H.,** Berres, A., Shao, Y., and Wang, C. (2022). Using Photo-Realistic Immersive Virtual Reality Digital Twin and Visualization to Support Road Safety Training, Transportation Research Board (TRB) - Visualization Symposium, National Academy of Sciences.
10. Berres, A., **Xu, H.,** Wang, C., Wang Q., Ugirumurera, J., Jones, W., and Sanyal, J. (2022). Multivariate Spatiotemporal Visual Representations for Analyzing Traffic Performance at Signalized Intersections, Transportation Research Board (TRB) - Visualization Symposium, National Academy of Sciences.
11. **Xu, H.,** Liu, X., Lian, J., Li, Y., Malhotra, M., & Omitaomu, O. A. (2022, November). A geo-visual analysis for exploring the socioeconomic benefits of the heating electrification using geothermal energy. In Proceedings of the 5th ACM SIGSPATIAL International Workshop on Advances in Resilient and Intelligent Cities (pp. 11-15).
12. **Xu, H.,** Yuan, J., Wang, C., Shao, Y., Berres, A., and Laclair, T. (2022). A Mobile App for Intersectional Traffic Optimization through Real-Time Vehicle-to-Infrastructure (V2I) Communication and Cyber-Physical Control. IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS 2022). Denver, CO.
13. Chinthavali, S., Hasan, S.M., Yoginath, S., **Xu, H.,** Nugent, P., Jones, T., Engebretsen, C., Olatt, J., Tansakul, V., Christopher Carter., Polsky An, Y., (2022). Alternative Timing and Synchronization Approach for Situational Awareness and Predictive Analytics, IEEE IRI 2022, Las Vegas, NV, USA. Resolution.
14. **Xu, H.,** Wang, C., Berres, A., Shao, Y., Yoshioka, J. (2022). Digital Twin and Real-time Situational Awareness for Sustainable and Smart Campus Management, AAG Annual Meeting 2022, New York, NY.
15. **Xu, H.,** Berres, A., Wang, C., Laclair, T. (2021). An Urban Informatics-based Vehicle-to-Infrastructure Communication Approach for the Smart Mobility and Environmental Management, AGU Fall Meeting, New Orleans, LA.
16. **Xu, H.,** Berres, A., Wang., C., LaClair, T., and Sanyal, J. (2021). Visualizing Vehicle Acceleration and Braking Energy at Intersections along a Major Traffic Corridor, ACM 2nd Workshop on Energy Data Visualization (EnergyVis II).
17. **Xu, H.,** Wang, C., Berres, A, LaClair, T. and Sanyal, J. (2021). RyThMiCCS: An Interactive Web-based Content Management System for Sharing and Comparing Traffic Simulation Outputs, Transportation Research Board Annual Meeting 2021.
18. **Xu, H.,** Sanyal., J., Berres, A. and Liu, Y. (2020). An Overview of Visual Analytics Applications in Water Resources Management, AWRA Annual Meeting, 2020.
19. Thakur, G., Sparks, K., Berres, A., Tansakul, V., Chinthavali, S., Whitehead, M., Schmidt, E., **Xu, H.,** Fan, J., Spears, D. and Cranfill, E., (2020), November. COVID-19 Joint Pandemic Modeling and Analysis Platform. In Proceedings of the 1st ACM SIGSPATIAL International

Workshop on Modeling and Understanding the Spread of COVID-19 (pp. 43-52).

20. **Xu, H.,** Sanyal., J., Berres, A. and Tennille, S. (2019). Optimization of Network Datasets for Web-based System using Composite Bezier Curves, AAG Annual Meeting 2020, Denver, Co.
21. Ugirumurera, J., Severino, J.A., Wang, Q., Ravulaparthy, S., Berres, A., Nugent, P., Sorensen, H., Moore, A., Todd, A., Nag, A. and Tennille, S., Peterson, S., Potter., K., **Xu, H.,** Jones. W., Sanyal., J. (2020). High Performance Computing Traffic Simulations for Real-Time Traffic Control of Mobility in Chattanooga Region (No. NREL/PO-2C00-75009). National Renewable Energy Lab.(NREL), Golden, CO.
22. **Xu, H.,** Berres, A. and Sanyal., J. (2019). A Client-side Web Application for Visualizing Massive Regional Mobility Data Collected from Real-time Traffic Sensors, AGU Fall Meeting 2019, San Francisco, CA
23. Muste, M., **Xu, H.** and Mocanu, M. (2017). Community Engagement in Water Resources Planning Using Serious Gaming, IEEE 13th International Conference on Intelligent Computer Communication and Processing," September 7-9, 2017, Cluj-Napoca, Romania.
24. **Xu, H.,** Hameed, H., Windsor, M., Muste, M., Demir, I., Smith, J., Hunemuller, T., Stevenson, M. B. (2017). Decision-support System for Underpinning Collaborative Planning for Multi-hazard Mitigation," Proceeding of the 37th IAHR World Congress, August 13-18, 2017, Kuala Lumpur, Malaysia.
25. **Xu, H.,** Hameed, H., Windsor, Smith, J., Hunemuller, T., M., Muste, M., Demir, I., Stevenson, M. B. (2017). An Iowa Prototype for a Generic Watershed Decision Support System, 11th Annual Iowa Water Conference, March 22-23, 2017, Ames, IA.
26. Muste, M. **Xu, H.,** Goetz, V., Claman, D., Abu-Hawash, A. (2017). Sedimentation Mitigation at Multi-box Culverts, 96th Annual Meeting, Transportation Research Board, January 8-12, 2017, Washington D.C.
27. Garousi-Nejad. I., Li. Xiao., Petrasova. A., and **Xu, H.** (2016). High-resolution Flood Inundation Mapping Using National Water Model Forecasts for Emergency Management, UCGIS symposium.
28. **Xu, H.,** Hameed, H., Demir, I., Muste, M. (2016). Visualization Platform for Collaborative Modelling, 2016 AWRA Summer Specialty Conference - GIS and Water Resources IX, American Water Resources Association, July 11-13, 2016, Sacramento, CA. –Awarded with the Best Student Oral Presentation for Haowen Xu.
29. **Xu, H.,** Hameed, H., Shen, B., Demir, I., Muste, M., Stevenson, M.B. and Hunemuller, T. (2016). Prototype Decision Support System for Interjurisdictional Collaboration in Water Resource Management, August 21-26, 2016, Hydroinformatics Conference, Incheon, Korea.
30. Muste, M., Cheng, Z., Firoozfar, A.R., Tsai, H-W., Loeser, T. and **Xu, H.** (2016). Impacts of Unsteady Flows on Monitoring Stream Flows, River Flow Conference, July 12-15, 2016, St. Louis, MO, USA.
31. **Xu, H.,** Shen, B. and Muste, M. (2015). Geo-portal for Sustainable Culvert Design and Monitoring, Proceedings 36th IAHR World Congress, 28 June – 3 July, The Hague, The Netherlands.
32. **Xu, H.,** Amir, A.A., Muste, M. (2014). Web-based Engine for Discovery of Observations Using Landscape Units," 11th International Conference on Hydroinformatics, New York City, NY.

TECHNICAL REPORTS AND WHITE PAPERS

1. **Xu, H.,** Berres, A. S., Yoginath, S., Kurte, K, Paleti, R., New, J., Sanyal. J. (2022). Towards Adaptive Decision Support: A Perspective from Intelligent and Annotated Visual Analytics for Exploring Big Urban Mobility Data, ASCR Workshop on Visualization for Scientific Discovery, Decision-Making, & Communication. U.S. Department of Energy.
2. **Xu, H.,** Berres, A. S., Yoginath, S., Sanyal. J. (2022). Metaverse for Urban Informatics: A Solution towards Participatory Smart City Applications, ASCR Workshop on the Science of Scientific-Software Development and Use. US Department of Energy.
3. Muste, M. and **Xu, H.** (2020). Development of self-cleaning box culverts, IIHR Report No. TR-719, Submitted to the Iowa Highway Research Board, Ames, Iowa.
4. Muste, M. and **Xu, H.** (2017). Mitigation of Sedimentation at Multi-Box Culverts, IIHR Report No. TR-655, Submitted to the Iowa Highway Research Board, Ames, Iowa.
5. Muste, M. and **Xu, H.** (2017). Sedimentation Mitigation using Streamlined Culvert Geometry, Report ST-001, Iowa Department of Transportation, Statewide Transportation Innovation Council, Federal Highway Administration, McLean, VA.
6. Riedinger, M., Ceynar, D., Eichinger, W., Muste, M., Plenner, S., Tsai, H.W., Willis, W. and **Xu, H.,** (2015). Salt Spray Measurement by Lidar, IIHR Report No. 482, Submitted to the Federal Highway Administration, Washington, D.C.

PROFESSIONAL CERTIFICATES

- IBM Professional Certificate - Machine Learning with Python (R96QPR685KBM)
- IBM Professional Certificate - Introduction to Deep Learning & Neural Networks with Keras (T43PD6PEWMJE)

PROFESSIONAL SERVICES

Guest Editor of Special Journal Issue

- **Topic Editor**, Topic "Agentic and Generative AI for Spatial Data Science: Toward Intelligent, Interactive, and Predictive Urban Systems", *ISPRS International Journal of Geo-Information*, MDPI, 2026
  https://x.com/ISPRS_IJGI/status/2064671982273069363
- Journal Special Issue "Frontiers in Urban Water Infrastructure", WATER, 2021
  https://www.mdpi.com/journal/water/special_issues/urban_water_infrastructure

Student Advisory & Mentorship

- Sidney Ozcan (Undergraduate) Co-advised through 2023 U.S. Department of Energy - Science Undergraduate Laboratory Internships (SULI) program
  Research Outcome: *Earthquake Vulnerability Assessment of California 's Petroleum Distribution Network (ORNL SULI Poster)*
- Jill Sturtevant (Ph.D Student at Baylor University) through 2023 NSF Internships Program
  Research Topic: *Understanding Traffic in and around Yellowstone National Park after the June 2022 Flooding*
- Jose Tupayachi (Ph.D Student at the University of Tennessee) through the ARPA-E RECOIL Project

Academic Services

- Grant Awardee and Exhibator, U.S. Department of Energy, ARPA-E Energy Innovation Summit 2024, Dallas, Texas, May 22-24, 2024
- Program Committee Member, ISPRS Technical Commission IV Symposium 2024

- Workshop Co-Organizer, IWCTS 2023 - The 16th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2023), Hamburg Germany, November 13, 2023
- Program Committee Member, Visualization in Environmental Sciences Workshop (EnvirVis) - EUROVIS 2023 - 25th EG Conference on Visualization, Leipzig, Germany, June 12-13, 2023
- Workshop Co-Organizer, IWCTS 2022 - The 15th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2022), Seattle, WA, November 1, 2022
- Program Committee Member, ARIC 2022 - The 5th ACM SIGSPATIAL International Workshop on Advances in Resilient and Intelligent Cities, Seattle, WA, USA, November 1, 2022
- Session Chair - Session "Geographic Information Analysis and Computing Technologies in Ecosystem Science" at AAG Annual Meeting 2022
- Session Co-organizer - Session "Symposium on Human Dynamics Research: Emerging Urban-Geo Big Data in Transportation Research" at AAG Annual Meeting 2022
- Program Committee Member, IWCTS 2021 - The 14th ACM SIGSPATIAL International Workshop on Computational Transportation Science (IWCTS 2021), Beijing, China, November 1, 2021
- Session Chair - Session "Geovisualization and Geovisual Analytics for Exploring Complex Urban Systems" at AAG Annual Meeting 2021
- Session Co-organizer - Session "Symposium on Human Dynamics Research: Emerging Urban-Geo Big Data in Transportation Research" at AAG Annual Meeting 2021

Referee Services

- **IEEE Transactions on Intelligent Transportation Systems**
- **Accident Analysis & Prevention**
- **Transportation Research Record**
- **Nature Cities**
- **Geo-spatial Information Science**
- **European Journal of Soil Science**
- **CATENA**
- **Computational Urban Science**
- **IEEE Computer Graphics and Applications**
- **Frontiers in Water**
- **Journal of Hydroinformatics**
- **Energies**
- **Sustainability**
- **Applied Sciences**

- **Open Research Europe**
- **Social Sciences**